

離線烹飪助理

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動機

- 隨著近年人工智慧技術的快速發展，AI 正逐漸融入日常生活之中。特別是 NPU (Neural Processing Unit) 的普及，使得 AI 能夠在 IoT 智慧家電上以更低功耗、更高效率的方式運作，讓「在地端執行 AI」成為可行的應用方向。基於這樣的趨勢，我希望探索大語言模型與 NPU 的結合應用。因此，本題以「廚房家電」為情境，設計並構想一款能夠完全離線運作的 AI 烹飪助理。透過本地推論的方式，使用者即使沒有網路，也能獲得即時的料理建議、步驟指引與互動式協助，展現未來智慧家電的可能性。

Neural Processing Unit - NPU

- 是專門為 AI 推理運算設計的處理器，特別擅長深度學習中的矩陣與向量計算。
- 低功耗、高效率
- 特別適合IoT、手機等設備

Retrieval-Augmented Generation - RAG

- RAG (檢索增強生成) 是一種讓大語言模型在回答問題時，能同時「查資料 + 生成答案」的技術。
- 它能降低幻覺、提升準確度。
- 知識可更新

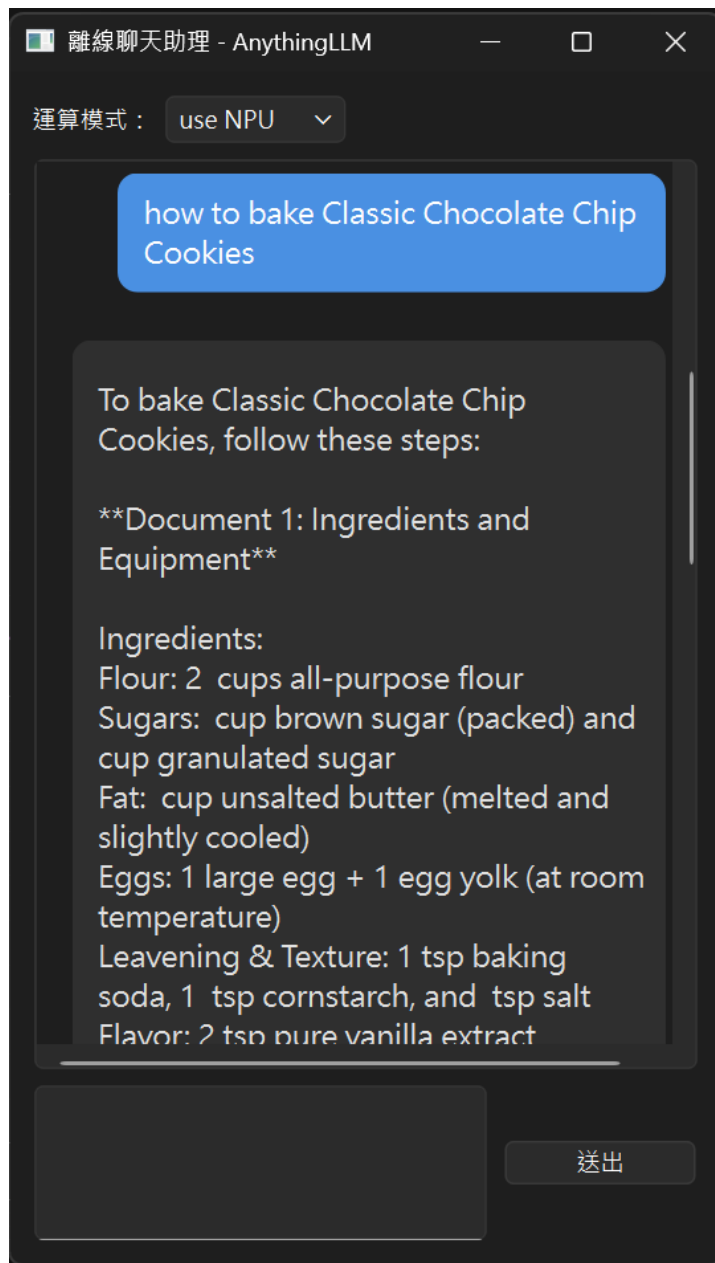
測試平台

- CPU : Qualcomm Snapdragon X Elite
- RAM : 16GB
- NPU : Qualcomm Hexagon NPU
- Model : Llama 3.2 3B
- Vector Database : LanceDB

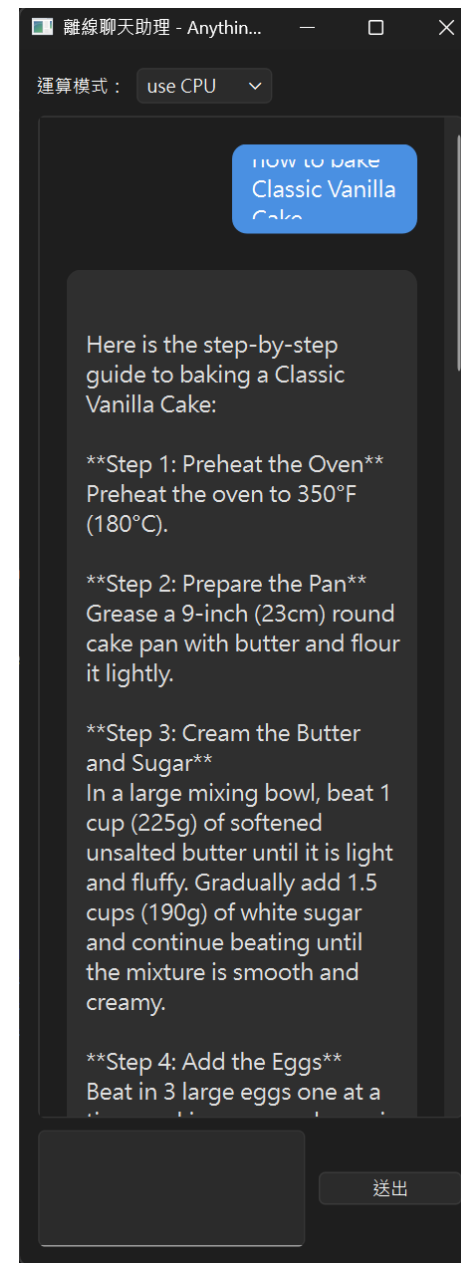


程式演示

使用NPU



使用CPU



程式演示

LLM後台

C:\Users\tonyt\AppData\Roaming\anythingllm-desktop\storage\models\QNN\llama_v3_2_3b_chat_8k

how to bake Classic Vanilla Cake

Here is the step-by-step guide to baking a Classic Vanilla Cake:

Step 1: Preheat the Oven Preheat the oven to 350°F (180°C).

Step 2: Prepare the Pan Grease a 9-inch (23cm) round cake pan with butter and flour it lightly.

Step 3: Cream the Butter and Sugar In a large mixing bowl, beat 1 cup (225g) of softened unsalted butter until it is light and fluffy. Gradually add 1.5 cups (190g) of white sugar and continue beating until the mixture is smooth and creamy.

Step 4: Add the Eggs Beat in 3 large eggs one at a time, making sure each egg is fully incorporated before adding the next.

Step 5: Add the Vanilla Extract Beat in 1 teaspoon of vanilla extract to enhance the flavor.

Step 6: Sift the Dry Ingredients In another bowl, sift together 2.5 cups (310g) of all-purpose flour, 2.5 teaspoons of baking powder, and 1 teaspoon of salt.

Step 7: Combine the Wet and Dry Ingredients Add the sifted dry ingredients to the butter mixture and mix until just combined.

Step 8: Add the Milk Alternate adding 1.25 cups (315ml) of milk and a small amount of the dry ingredients until the batter is smooth.

Step 9: Pour the Batter into the Pan Pour the batter into the prepared pan and smooth the top.

Step 10: Bake the Cake Bake the cake for 25–35 minutes, or until a toothpick inserted into the center comes out clean.

Step 11: Cool the Cake Let the cake cool in the pan for 5–10 minutes, then remove it to a wire rack to cool completely.

Tips and Variations

Send a message

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Sources

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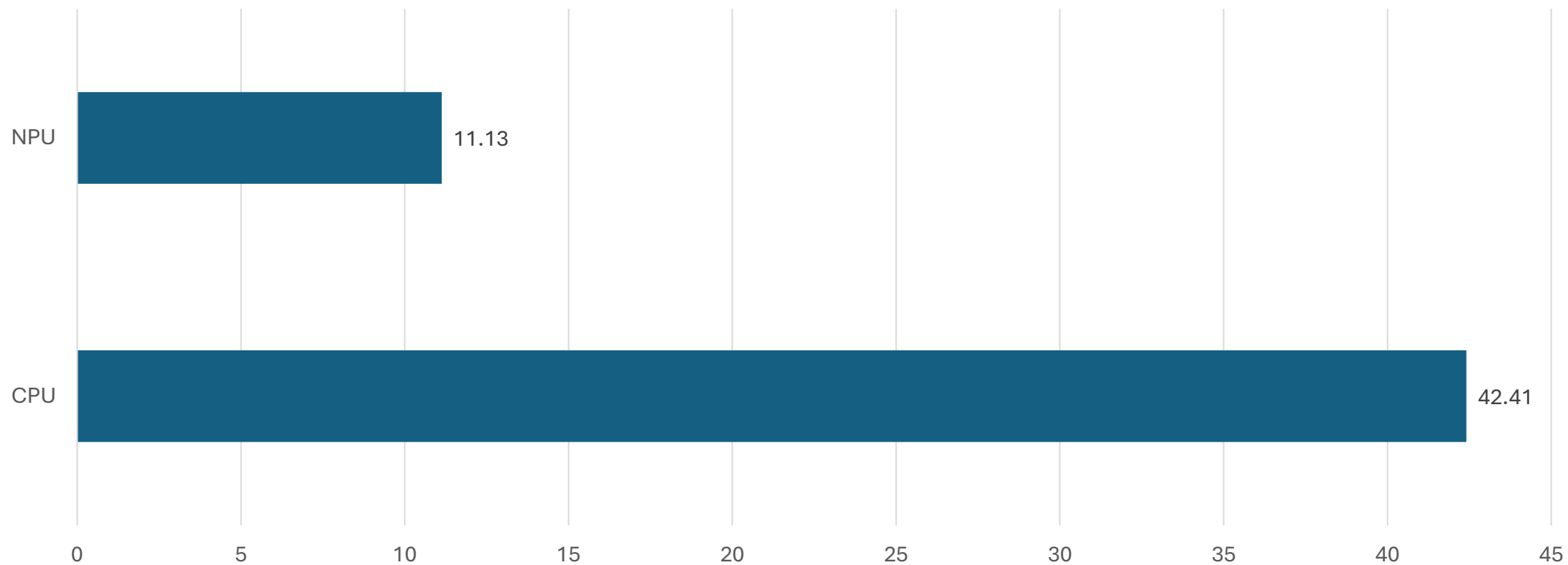
📄 Classic Vanilla Cake Recipe.txt
Document
2 references

📄 Classic Sponge Cake Recipe.txt
Document
1 reference

📄 Classic Chocolate Chip Cookies.txt
Document
1 reference

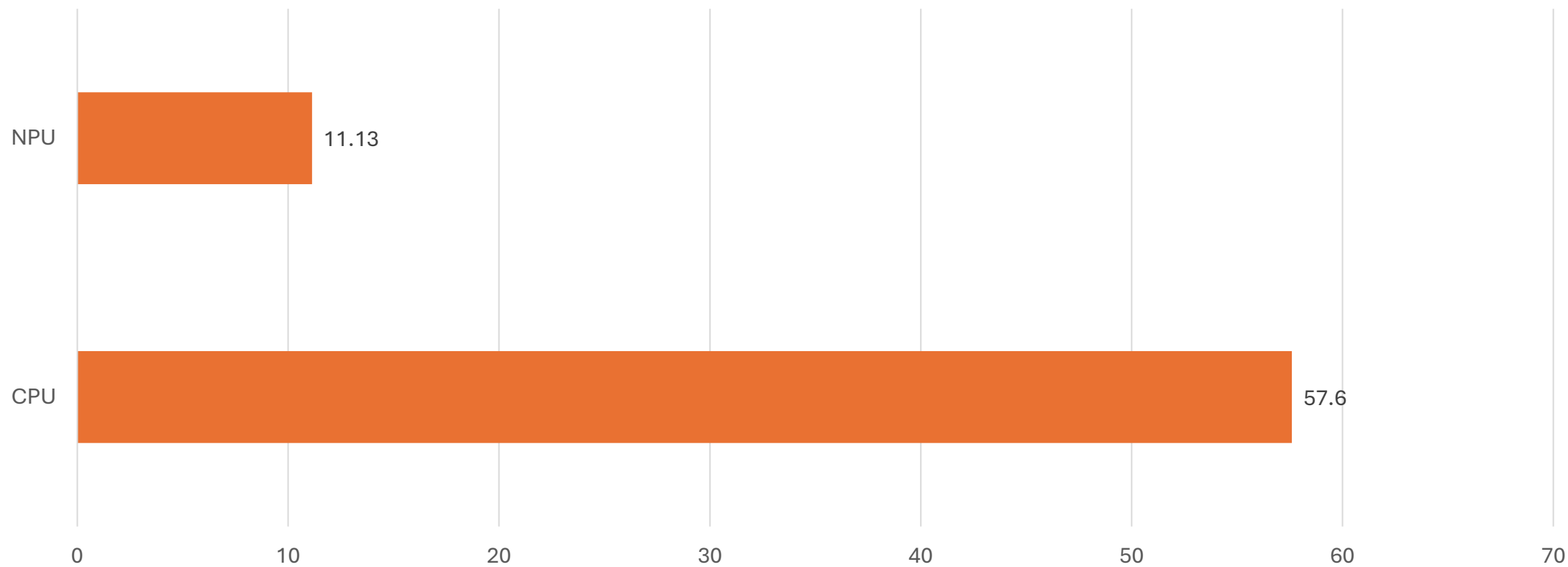
效能

每秒Token數 (tok/s)

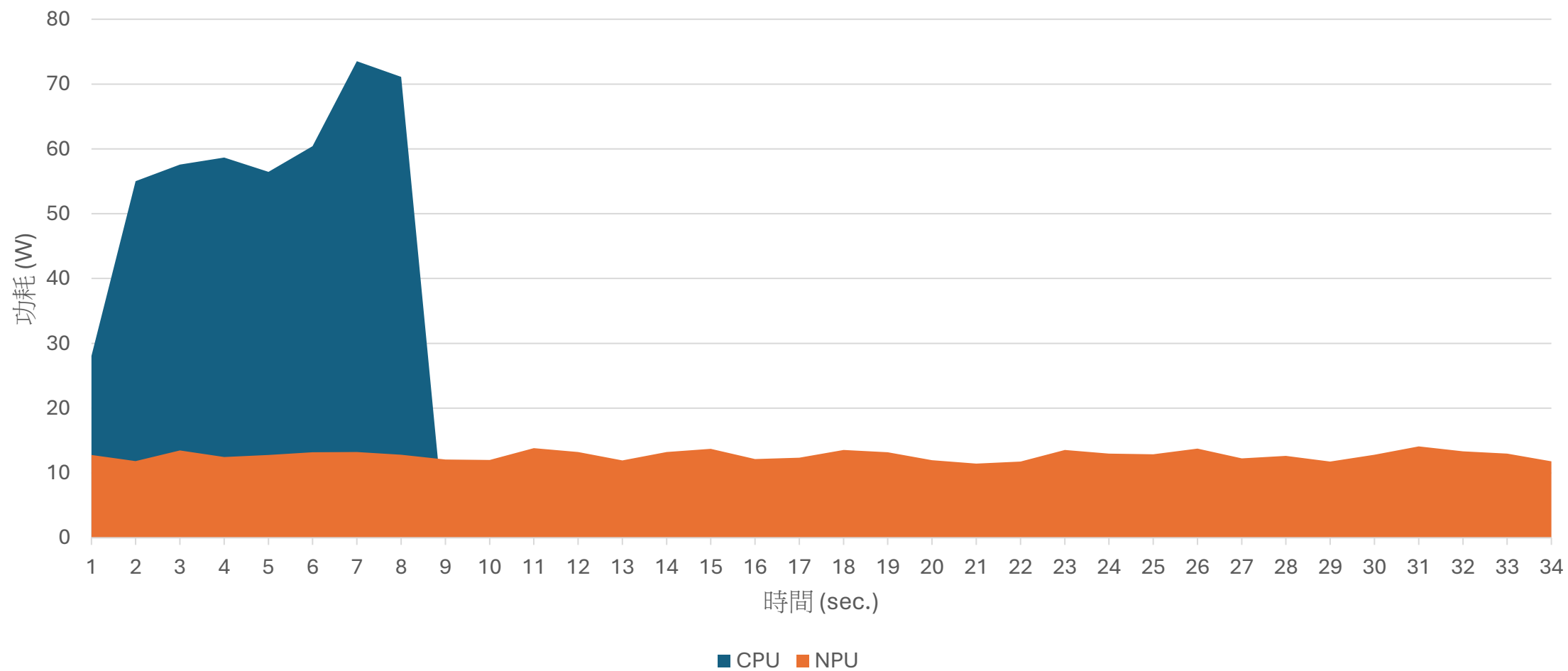


功耗

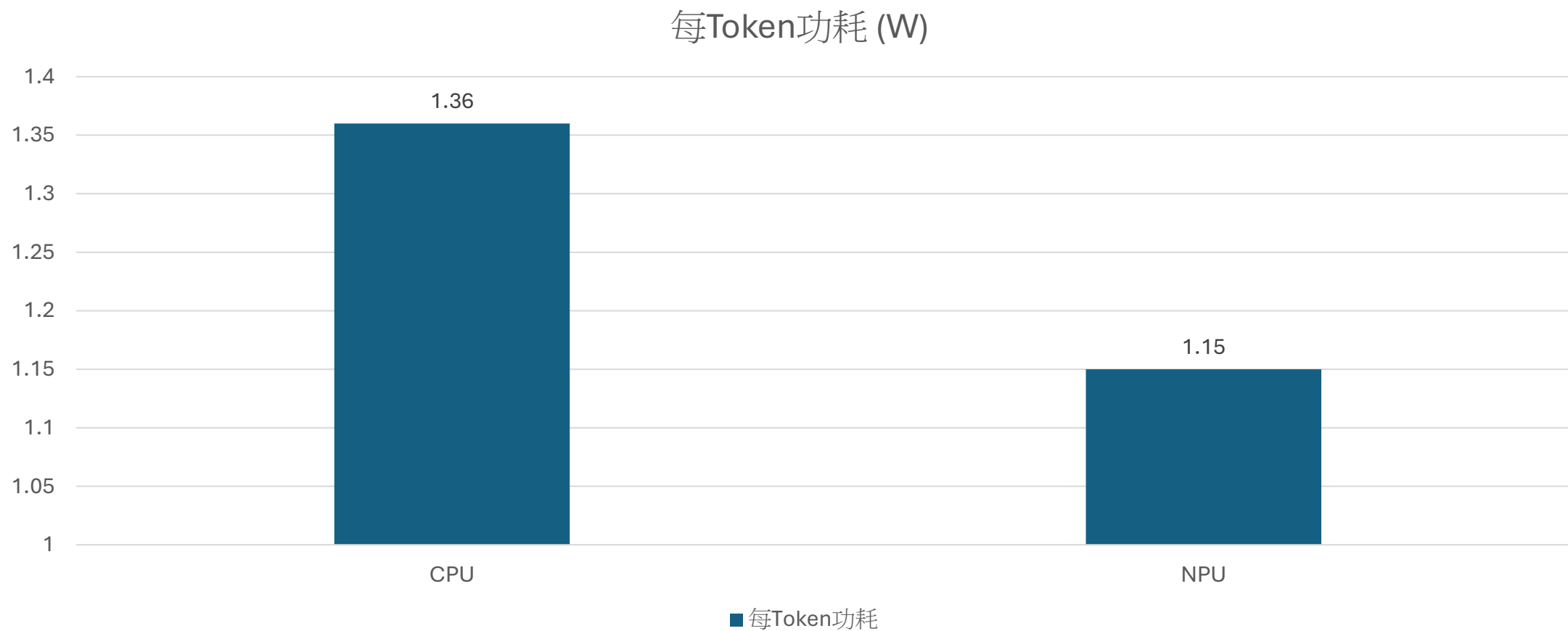
平均功耗 (W)



功耗紀錄



每Token的功耗



結論

- 由以上的測試可以得知，使用NPU進行推理加速，雖然絕對的速度不及CPU運算，但能有效提高運算效率和降低功耗，非常適合在IoT等設備實裝。